

Prediction of disulfide connectivity in proteins

Bioinformatics Vol. 17 no. 10 2001
Pages 957-964

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Date: Dec. 3, 2008

Abstract (1/3)

- Motivation: A major problem in protein structure prediction is the correct location of disulfide bridges in cysteine-rich proteins. In protein-folding prediction, the location of disulfide bridges can strongly reduce the search in the conformational space. Therefore the correct prediction of the disulfide connectivity starting from the protein residue sequence may also help in predicting its 3D structure.

Abstract (2/3)

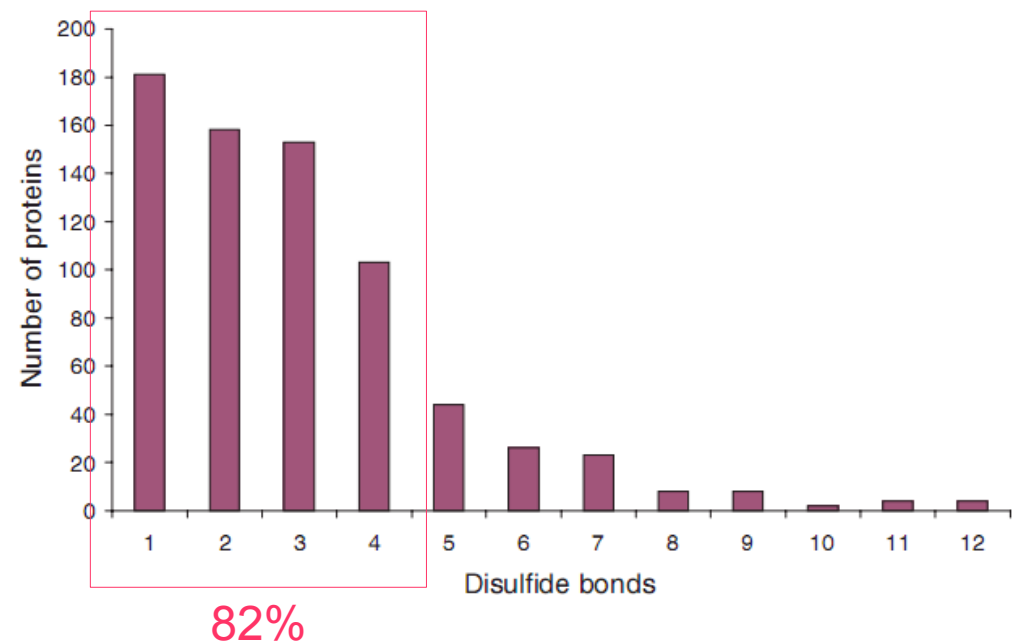
- Results: In this paper we equate the problem of predicting the disulfide connectivity in proteins to a problem of finding the graph matching with the maximum weight. The graph vertices are the residues of cysteine-forming disulfide bridges, and the weight edges are contact potentials. In order to solve this problem we develop and test different residue contact potentials.

Abstract (3/3)

- The best performing one, based on the Edmonds–Gabow algorithm and Monte-Carlo simulated annealing reaches an accuracy significantly higher than that obtained with a general mean force contact potential. Significantly, in the case of proteins with four disulfide bonds in the structure, the accuracy is 17 times higher than that of a random predictor. The method presented here can be used to locate putative disulfide bridges in protein-folding.

Data Set

- 726 proteins from SWISS-PROT NO.39
 - Remove *probable*, *potential* or *by similarity*
 - Grouped 4 set with sequence homology among the other set was $\leq 30\%$
 - 4-fold cross validation



Problem definition

- Performance criteria:

- $Q_p = \delta(\text{correct pattern}, \text{predicted pattern})$
- $Q_c = \frac{\text{number of correctly precivted pair}}{\text{number of possible pairs}}$

- Random predictor:

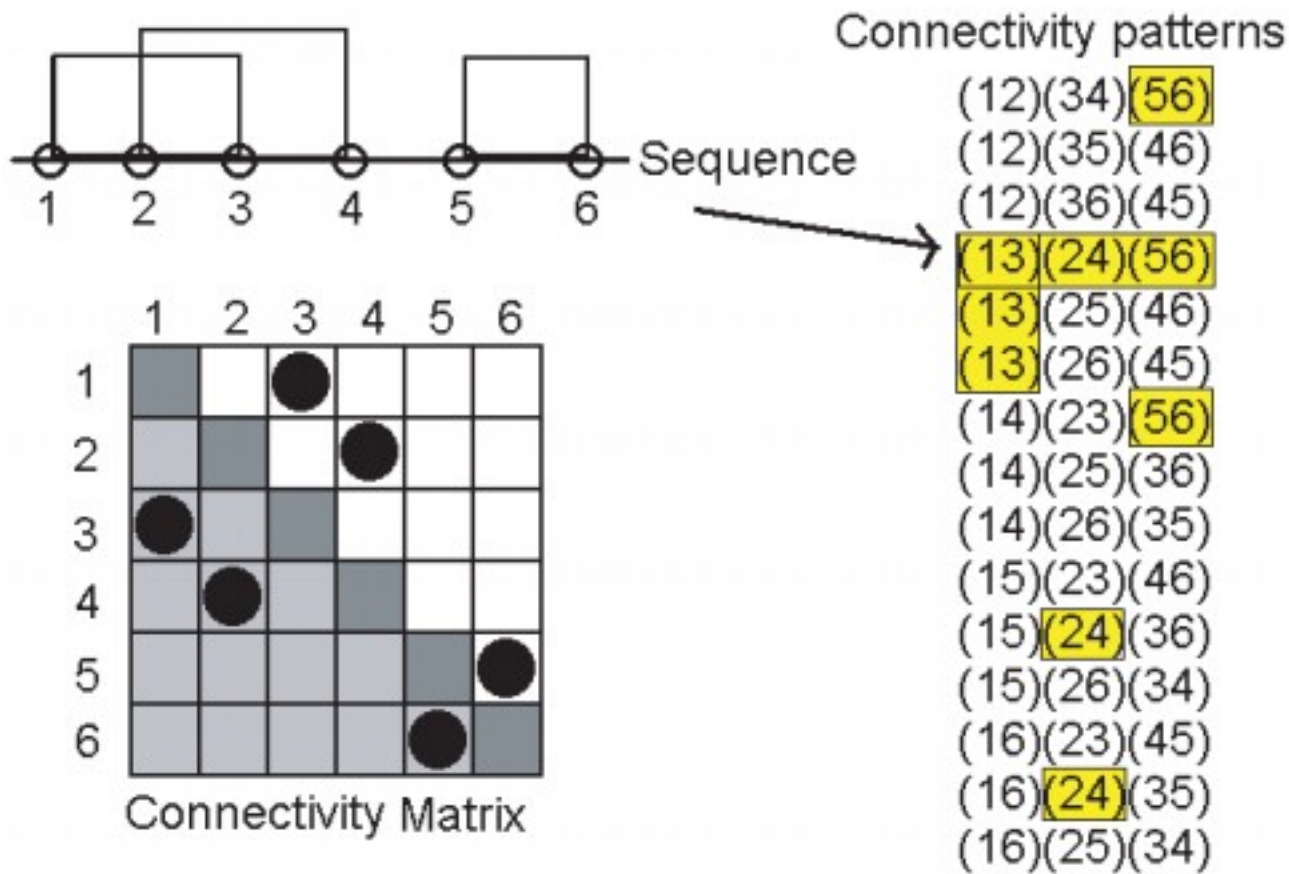
- Possible connectivity patterns: $N_p = (2B-1)!!$

B	2	3	4	5	6
N_p	3	15	105	945	10395

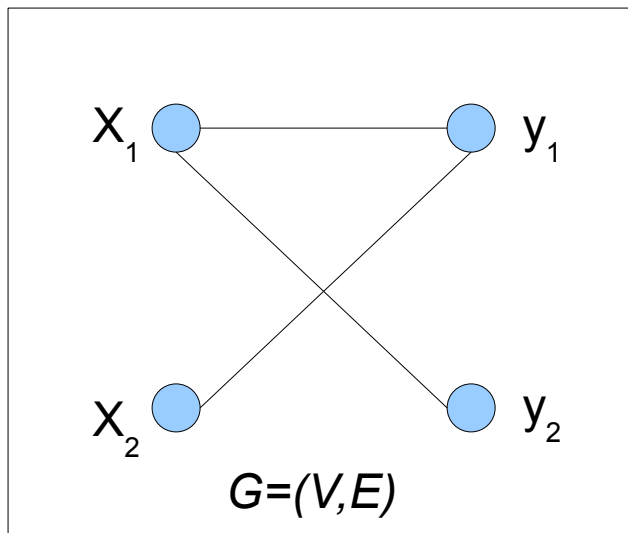
- $Q_p(R_p) = 1/N_p$
- $Q_c(R_p) = 1/(2B-1)$

Subject

- Maximum-weight perfect matching



Graph matching



$M = \{\{x_1, y_1\}\}$ is a maximal matching, but not a maximum matching.

$M' = \{\{x_1, y_2\}, \{x_2, y_1\}\}$ is a maximal matching and also a maximum matching.

- A maximal matching is a matching M of a graph G with the property that if any edge not in M is added to M , it is no longer a matching.
- A maximum matching is a matching that contains the largest possible number of edges.
- A perfect matching is a matching which matches all vertices of the graph. Each vertex is met by only one edge in the set.

Preliminary work

- Window size: 11 residues, center on cysteine

TABLE I. Performance of the Predictor Using Single-Sequence-Based Input

Window length	Q3 (%) ^a	C	Pc (SS)	Pc (SH)
7	69.3 (0.9)	.33	.63	.72
9	70.4 (0.9)	.38	.64	.73
11	71.4 (0.9)	.40	.66	.73
13	71.8 (0.9)	.41	.67	.74
15	71.1 (0.9)	.40	.66	.73
17	70.0 (0.9)	.35	.60	.75

TABLE II. Performance of the Predictor Using Multiple-Sequence Alignment-Based Input

Window length	Q3 (%) ^a	C	Pc (SS)	Pc (SH)
7	74.9 (0.9)	.43	.65	.79
9	76.5 (0.9)	.47	.67	.81
11	77.7 (0.8)	.50	.68	.82
13	77.5 (0.8)	.50	.68	.82
15	77.9 (0.8)	.50	.69	.82
17	78.2 (0.8)	.51	.70	.82

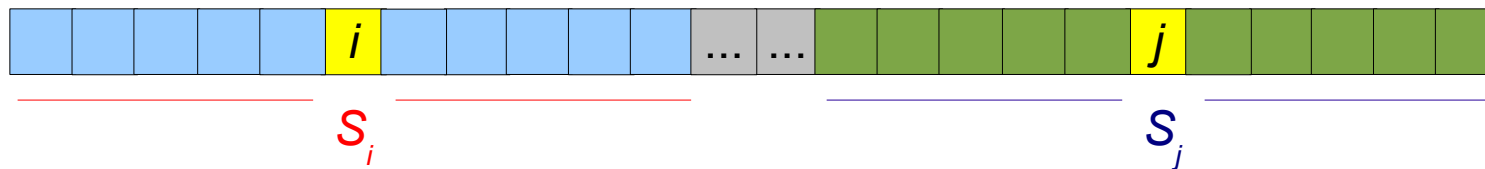
Role of Evolutionary Information in Predicting the Disulfide-Bonding State of Cysteine in Proteins [Fariselli et. al., 1999]

- Captured features:
 - Protein metal binding cysteines are typically found in position i and $i+3$
 - Hydrophilic and/or charged residue in the environment are highly conducive towards disulfide bond formation

Weight edge

- Assumption: for each cysteine, all its sequence nearest neighbors make contacts.
- Weight of connection between i and j

$$w_{i,j} = \sum_{k \in S_i} \sum_{l \in S_j} U(k,l) \quad , \text{ where } U \text{ is the residue contact potential}$$



- Constraint:
 - $\langle U \rangle = 0$, average interaction between amino acid
 - $\sigma^2(U) = 1$, the dispersion of the interaction energy

Result (1/5)

- With MS contact potential

No. of chains	B	$Q_p(MS)$	$Q_p(MS)/Q_p(R_p)$	$Q_c(MS)$	$Q_c(MS)/Q_c(R_p)$
158	2	0.34	1.0	0.35	1.0
153	3	0.05	0.9	0.19	1.0
103	4	0.009	1.0	0.14	1.0
44	5	0.0	0.0	0.12	1.0

Result (2/5)

- With constrained optimization contact potential

No. of chains	B	$Q_p(CO)$	$Q_p(CO)/Q_p(R_p)$	$Q_c(CO)$	$Q_c(CO)/Q_c(R_p)$
158	2	0.42	1.2	0.42	1.2
153	3	0.09	1.3	0.21	1.0
103	4	0.05	5.0	0.21	1.5
44	5	0.0	0.0	0.14	1.3

$$U(i, j) = [A(i, j) - \langle A(i, j) \rangle] / \sigma^2(A(i, j))$$

Result (3/5)

- With odd-ratio contact potential

No. of chains	B	$Q_p(OR)$	$Q_p(OR)/Q_p(R_p)$	$Q_c(OR)$	$Q_c(OR)/Q_c(R_p)$
158	2	0.46	1.4	0.46	1.4
153	3	0.17	2.4	0.29	1.4
103	4	0.11	11.5	0.31	2.2
44	5	0.0	0.0	0.17	1.5

$$U(i, j) = \frac{P(i, j|T)}{P(i, j|N)}$$

Space: $C(20,2)=210$

Result (4/5)

- With EG algorithm

No. of chains	B	$Q_p(EG)$	$Q_p(EG)/Q_p(R_p)$	$Q_c(EG)$	$Q_c(EG)/Q_c(R_p)$
158	2	0.56	1.7	0.56	1.7
153	3	0.21	3.1	0.36	1.7
103	4	0.17	17.8	0.37	2.6
44	5	0.02	22.7	0.21	1.9

- Using odd-ratio as start value with Monte-Carlo simulated annealing

Result (5/5)

- The relevance of residue pair to form disulfide bond

